

DINETHRA DIVANJANA RAJAPAKSHA

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SUMMARY

Final-year undergraduate from the University of Moratuwa with a focus on machine learning, Computer Vision and software development supported by a strong background in embedded systems and IoT. Seeking opportunities to contribute to software and AI-driven engineering projects in innovative environments.

EXPERIENCE

Magicbits (PVT) Ltd / ThingsNode (PVT) Ltd

Dec 2024 – Jun 2025 — On-site

Software & IoT Engineering Intern

- Individually complete **ESP32 firmware development for Ansell**, implementing OTA update workflows, watchdog-based recovery, flash/NVS-based configuration storage, and **network-load optimization** strategies to improve field stability.
- Built a **fingerprint sensor-based user monitoring system for Ansell Company**, integrating ESP32 firmware with a UART fingerprint module, an on-device web interface (admin enrollment, user management), and MQTT telemetry to ThingsBoard for monitoring.
- Contributed to the **Ansell Company HMI project** by developing Lua scripts, register mappings, and device interaction logic for **WECON (PiStudio)** displays.
- Developed and tested **IoT device-cloud communication workflows** using MQTT, HTTP, and Modbus (RTU/TCP) across ThingsNode devices, focusing on reliability and production-readiness.
- Developed a **production monitoring dashboard for Quick Tea (Wattala)**, enabling real-time observation of device data, operational parameters, and system-level metrics.
- Customized and debugged the **ThingsNode mobile application** using Android Studio and Flutter, resolving build/configuration issues, adapting branding elements, and improving notification workflows via Firebase Cloud Messaging (FCM).

EDUCATION

University of Moratuwa, Moratuwa, Sri Lanka

July 2022 – Present

B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering GPA: 3.65 / 4.00

Dean's List: Semester 4, Semester 6, Semester 7

- *Relevant Coursework:* Data Structures and Algorithms, Applied Statistics, Signals and Systems, Machine Learning, Deep Learning for Vision, Image Processing and Machine Vision, Pattern Recognition, Neural Networks and Fuzzy Logic, Engineering Optimization, Autonomous Systems, Software Design Competition

Bandaranayake College, Gampaha, Sri Lanka.

January 2012 - October 2020

GCE Advanced Level - Physical Science Stream; Z-Score 2.5516. 

- *Combined Mathematics, Physics, Chemistry.*

GCE Ordinary Level ; 9As. 

PROJECTS

Quadruped Mobility System for Dynamic Sensing (Unitree Go2) — FYP – Ongoing (Ongoing)

- **Tools & Technologies:** Python, C++, CUDA, ROS 2 Humble, VLM, PyTorch, YOLOv11-OBB, YOLOv8, OpenCV, FastAPI, SQLite, Docker, Jetson Orin Nano, Unitree Go2, EasyOCR, PaddleOCR, scikit-learn, NumPy, TensorRT
- **Project Overview:** Designed a real-time visual inspection system for quadruped-mounted industrial monitoring, integrating GPU-accelerated object identification and understanding, multi-stage analog gauge reading, and distributed edge-server workflows for autonomous inspection.
- **My Contributions:** Full Visual Inspection system
 - Developed a real-time perception pipeline using **YOLO-based object detection and tracking**, optimized with **TensorRT** for low-latency inference on **Jetson Orin Nano**.

- Implemented a hybrid analog gauge reading framework combining **CNN-based detection**, **vision transformer keypoint regression**, geometric ellipse fitting, segmentation-based needle detection, OCR with rotation normalization, and **RANSAC-based angle-to-value regression**.
- Integrated vision-based anomaly identification to detect abnormal gauge states and inconsistent readings during inspection.
- Performed camera–LiDAR mapping and spatial alignment to associate detected visual assets with robot-centric spatial context for accurate inspection localization.
- Integrated zero-shot **VLM-based reasoning** with structured JSON parsing as a fallback pipeline, achieving 98.1% accuracy under unseen target objects types.
- Integrated a dual-camera setup (Insta360 wide-view + pan tilt Logitech camera), mapping wide-view detections to pan tilt commands via regression and closing the loop with **image-based visual servoing** to center targets in the Logitech stream.

AWS Cloud-Native Research & Documentation Chatbot (LLM + RAG + MLOps) (Ongoing)

- Designed and implemented an **AWS cloud-native LLM-based chatbot** for **research papers and technical documentation** using **Retrieval-Augmented Generation (RAG)**, enabling context-aware question answering over continuously expanding document collections.
- Built an automated ingestion and retrieval pipeline using **FastAPI**, **FAISS vector search**, and **local LLM inference via Ollama**, with **incremental indexing**, containerized deployment via **Docker**, and **MLOps-oriented project structure**.

TransX – Transformer Maintenance Platform

November 2025

- Developed a full-stack AI-powered platform for transformer inspection, integrating React/TypeScript frontend, Spring Boot backend, and a YOLOv8-based Python ML microservice for real-time thermal anomaly detection and interactive annotation workflows.
- Implemented the complete database schema (MySQL, relational database), REST API endpoints, and frontend modules for inspection management, annotation canvas, and maintenance record generation.

Event-Driven Workflow Management Platform

(Ongoing)

- Designed and implemented a full-stack event-driven workflow platform using Apache Kafka for asynchronous task processing, featuring REST API with Express, Kafka-based event streaming, and React frontend with real-time status updates demonstrating microservices architecture and distributed systems patterns.
- Independently developed all system components: backend API with event publishing, Kafka consumer worker with step-by-step processing logic, React UI with auto-refresh capabilities, Docker-based infrastructure setup, and comprehensive technical documentation including architecture design, API specifications, and event schemas.

YOLO & Kalman filter based Target Tracking System

December 2025

- Designed and implemented an end to end video object tracking system combining YOLO-based detection with Kalman filter multi object tracking for robust, real-time tracking by detection, including occlusion handling and jitter reduction.
- Independently developed all modules: dataset preparation, model training, inference pipelines, Kalman tracking logic, visualization utilities, and automated result analysis with annotated videos and performance plots.

FasterViT-OCI: Boundary-Aware Hierarchical Vision Transformer Extension

December 2025

- Implemented and evaluated the FasterViT architecture, extending the original paper with OCI improvements to enhance hierarchical attention mechanisms; conducted three comprehensive experiments: inference throughput replication on ImageNet-1K, transfer learning on CIFAR-10, and object detection benchmarks with Detectron2 backbones on Pascal VOC.
- Designed and executed the full training and evaluation pipeline for FasterViT-0 on CIFAR-10, including data preprocessing, augmentation, checkpointing, and metric logging. Implemented the OCI extension in PyTorch, trained and evaluated both baseline and OCI models, and performed quantitative and qualitative comparisons.

Vision-Based Object Tracking and Control System (Robotic Arm)

July 2025

- Developed a real-time vision-based system that detects and tracks a selected target object and generates continuous control commands to align and follow the target using a robotic arm.
- Implemented the complete perception-to-control pipeline, including camera-based object detection and tracking, coordinate mapping from image space to actuator commands, and embedded control logic for smooth and stable object-following behavior.

CLRerNet Improving Confidence of Lane Detection with LaneloU

August 2024

- CLRerNet enhances lane detection confidence using the LaneloU architecture, improving accuracy in complex driving scenarios. The project focuses on advanced neural network techniques to refine lane recognition in real-time.
- Develop the CLRerNet algorithm using Python, leveraging the LaneloU architecture to improve lane detection confidence. The implementation will involve training and fine-tuning the neural network to enhance accuracy in detecting lanes under diverse driving conditions.

Hand Gesture Detection using machine learning and Image processing

June 2024

- This project harnesses the power of machine vision and deep learning to create a image recognition system.
- implemented a hand gesture detection system using CNNs, achieving high accuracy in gesture recognition. I optimized the model for real-time tracking and classification.

Ratatouille : Fast Maze Traversing Micro mouse

June 2024

- The Fast Maze-Traversing Micro mouse navigates complex mazes autonomously using advanced sensors, algorithms.
- Designed schematics and PCBs and working with Webots to simulate and optimize the micro mouse's performance.

TechbBot : STM32/Vision-Based Robot

March 2023

- An innovative robot featuring a custom-designed 4-layer PCB with an STM32F405VGT6 ARM-M4 microcontroller and a Raspberry Pi 4b for advanced machine vision tasks.
- Designed the shooter mechanism, worked with Cube IDE for development, tested sensors, assembled mechanical and electronic components, and was involved in algorithm design using C++.

Other Computer Vision & Machine Learning Projects

- Simple Convolutional Neural Network for Classification 
- Image Processing & Machine Vision 
- Intensity Transformations & Neighborhood Filtering 
- Binary Classification with Multi-Layer Perceptron (MLP) 
- RNN-based Word Autocompletion 
- MLP for MNIST Digit Classification 
- CNN for CIFAR-10 Image Classification 

Other Engineering Projects

- Smart Rolling Door Device 
- UART Transceiver Implementation in FPGA 
- Nanonaut: Task-Completing Line Following Robot 
- Path Planning in Robot Operating System (ROS2) 
- Turbi-Detector: IoT-Based Solution for Contaminated Water 
- IoT-Based Medicine Storage System 
- Industrial Smart Soldering Station 
- Puma560 Robot Simulation using MATLAB 
- Analog Guitar Headphone Amplifier 

AWARDS AND SCHOLARSHIPS

Finalist — Sri Lanka Robotic Challenge 2024



Competed in the prestigious Sri Lanka Robotic Challenge 2024, demonstrating advanced robotic design and programming skills. Achieved finalist status by successfully navigating through complex challenges and showcasing innovative solutions in the university category.

Mahapola Higher Education (Merit) Scholarship.

For outstanding performance in the GCE A/L Examination. Ranked #124 out of 34389 students.

TECHNICAL SKILLS AND INTERESTS

Programming: Python, C/C++, MATLAB, SQL, React.js

Developer Tools: Altium Designer, SolidWorks, LTspice, Multisim, Quartus Prime, Arduino, Atmel Studio, PIS-tudio, Webots, Version Control (Git and GitHub), SLAM toolbox, TensorFlow

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn

Cloud / Databases: SQL, Firebase

Soft Skills: Problem Solving, Decision Making, Project Management, Team Leadership, Marketing

Coursework: Machine Vision, Deep Learning, Applied Machine Learning, Engineering Project Management

Sports & Other: Badminton, Playing the Guitar

CERTIFICATIONS

Deep Learning & Machine Learning: Convolutional Neural Networks (DeepLearning.AI, 2025), Structuring Machine Learning Projects (DeepLearning.AI, 2025), Supervised Machine Learning: Regression and Classification (DeepLearning.AI, Stanford University, 2024)

Programming & Problem Solving: Problem Solving – Basic (HackerRank, 2024), Problem Solving – Intermediate (HackerRank, 2024), Python – Basic (HackerRank, 2024)

VOLUNTEER EXPERIENCE

University of Moratuwa

Career Fair 2023 — Company Coordinator (ADL, Axiata Digital Lab) • Sri Lanka Robotic Challenge 2023 — Video Streaming Coordinator • Abhina 2023 — Video Streaming Coordinator • Exmo Exhibition — Coordinator

Rotaract Club of University of Moratuwa

Jun 2022 – Apr 2023

General Member; Organizing Committee for Professional Development, Membership Development and Community Service projects

Bandaranayake College, Gampaha

Science Society — Coordinator (Xban Exhibition) • Scout Association — Member • Buddhist Society — Member • Xban Exhibition (2014, 2018) — Project Coordinator (Astronomy, Pyramid Technology)

REFERENCES

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